

Baselines First

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Calculate the validation score for A baseline before a model. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for A baseline before a model. Proposed explanation: “A simple baseline does not need training data”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Fit and compare independently

New training labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same baseline rule and list its four predictions. A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and percentage points. What claim remains unsupported?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

6. Solve the given case

A test set has eighteen healthy and two spotted leaves. Baseline always predicts healthy.

Calculate and critique accuracy.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A dataset has 18 grass and 2 fern examples. A baseline always predicts grass. Compute accuracy and name a serious limitation.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 10 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Record majority and transparent-rule baselines for the Plant Classifier using the frozen test plan.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy baseline gets what accuracy and spotted recall? What must a new model improve upon?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Choose from training evidence

Count clear and blocked in the worked training list. State the constant prediction and the declared tie rule. Explain why the validation majority does not choose this baseline.

My reasoning and evidence

13. W2 Calculate the validation score

List the four baseline predictions and mark which validation rows are correct. Give correct/total and accuracy. Explain why 4/6 is not the answer.

My reasoning and evidence

14. W3 Make the comparison fair

A candidate is scored on ten easy rows and the baseline on four difficult rows. Describe a fair comparison and two details you must record.

My reasoning and evidence

15. W4 Fit and compare independently

New training labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same baseline rule and list its four predictions. A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and percentage points. What claim remains unsupported?

My reasoning and evidence

16. W5 Explain the hidden error

A training-derived clear baseline gets nine of ten validation cases correct; the labels are nine clear and one blocked. State its accuracy and missed case. If a candidate also gets nine correct, must it miss the same case? Explain.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Baselines First

Make a simple baseline earn its place before claiming a better classifier.

Training labels: healthy, healthy, spotted. Majority baseline predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Baselines First

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit baseline

Training majority is healthy: 2 of 3 labels.



2. Score baseline

Always healthy matches 2/4 validation cases: 50%.



3. Compare candidate

Supplied candidate predictions healthy, spotted, spotted,
spotted match 3/4: 75%.

CHANGE ONE THING

Change the third candidate prediction from spotted to healthy.

Explain the changed result using the matching lesson.

Can the candidate's training accuracy replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Make a simple baseline earn its place before claiming a better classifier.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

1. Fit baseline
2. Score baseline
3. Compare candidate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare candidate Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%. Read the picture in order. Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Change the third candidate prediction from spotted to healthy. Which conclusion is supported by the supplied facts?

1. Candidate becomes correct on all 4 validation cases.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Candidate becomes correct on all 4 validation cases. Only the previously wrong third decision changes; baseline stays fixed at 2/4.

Repair a missing building block

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table schema. Short practice: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
